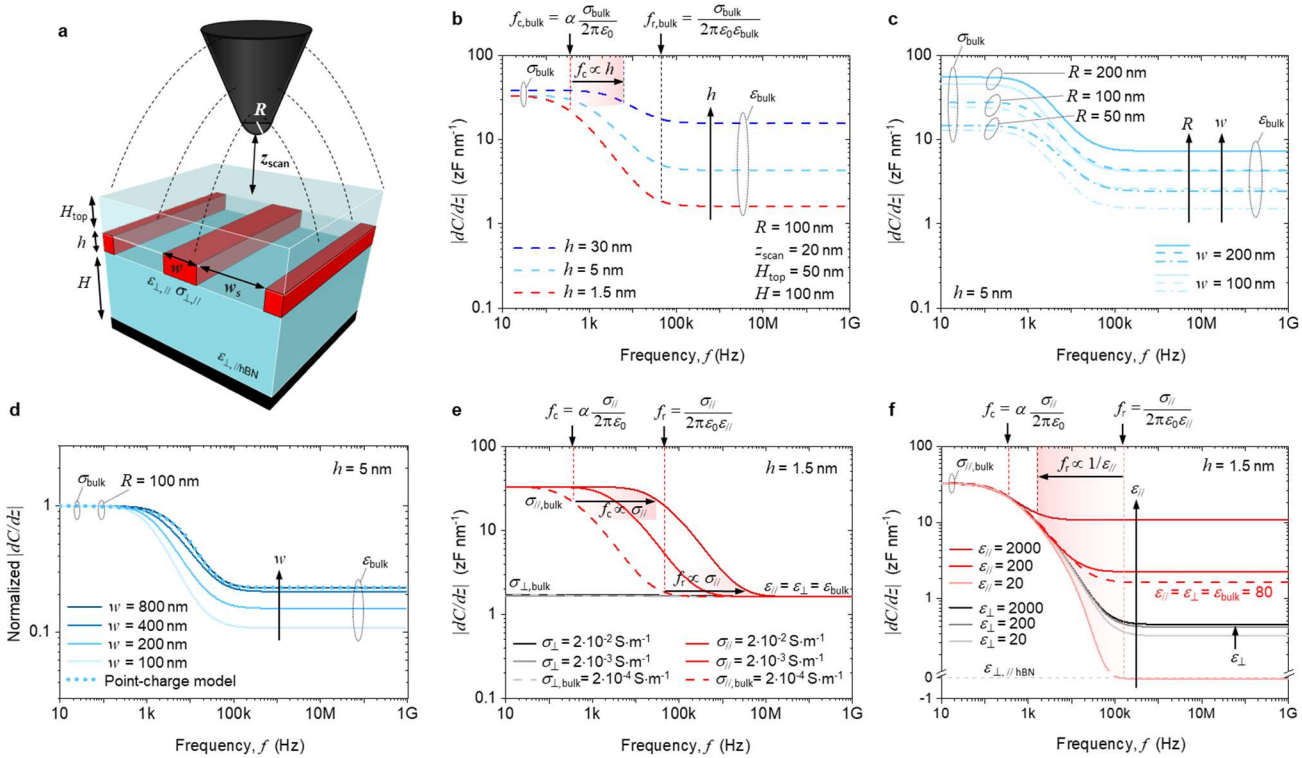
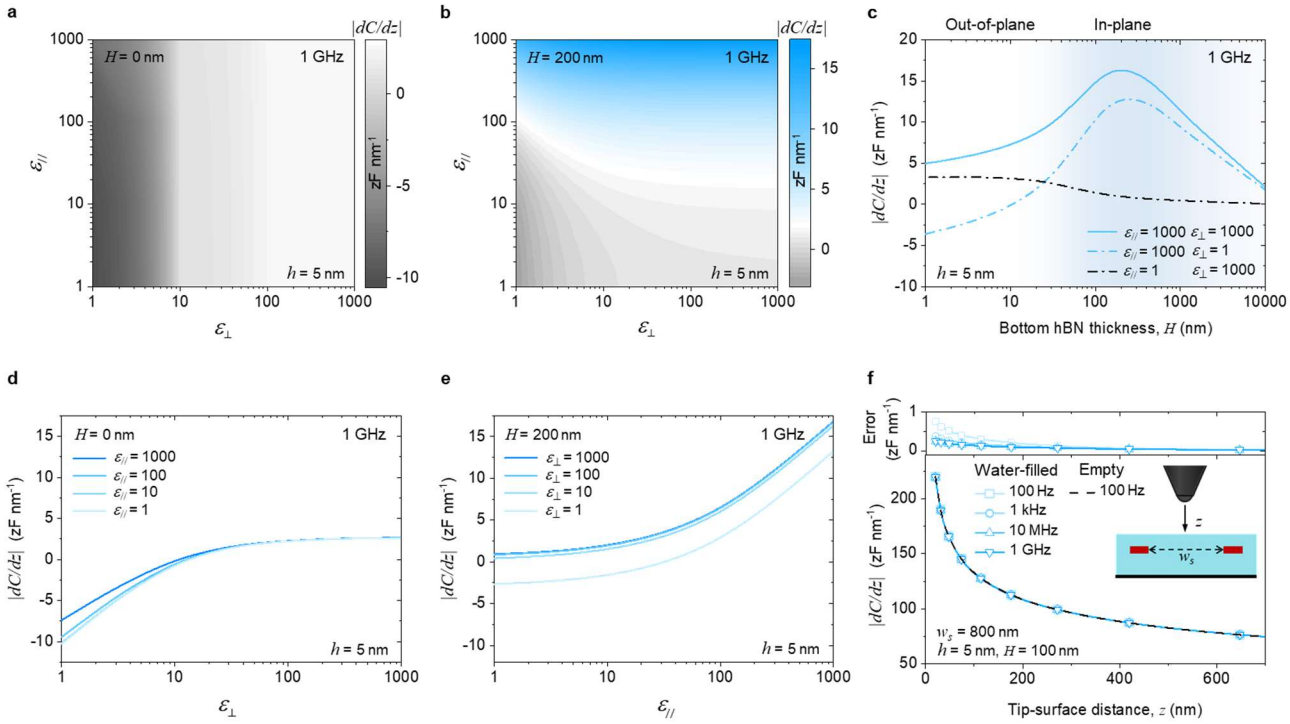
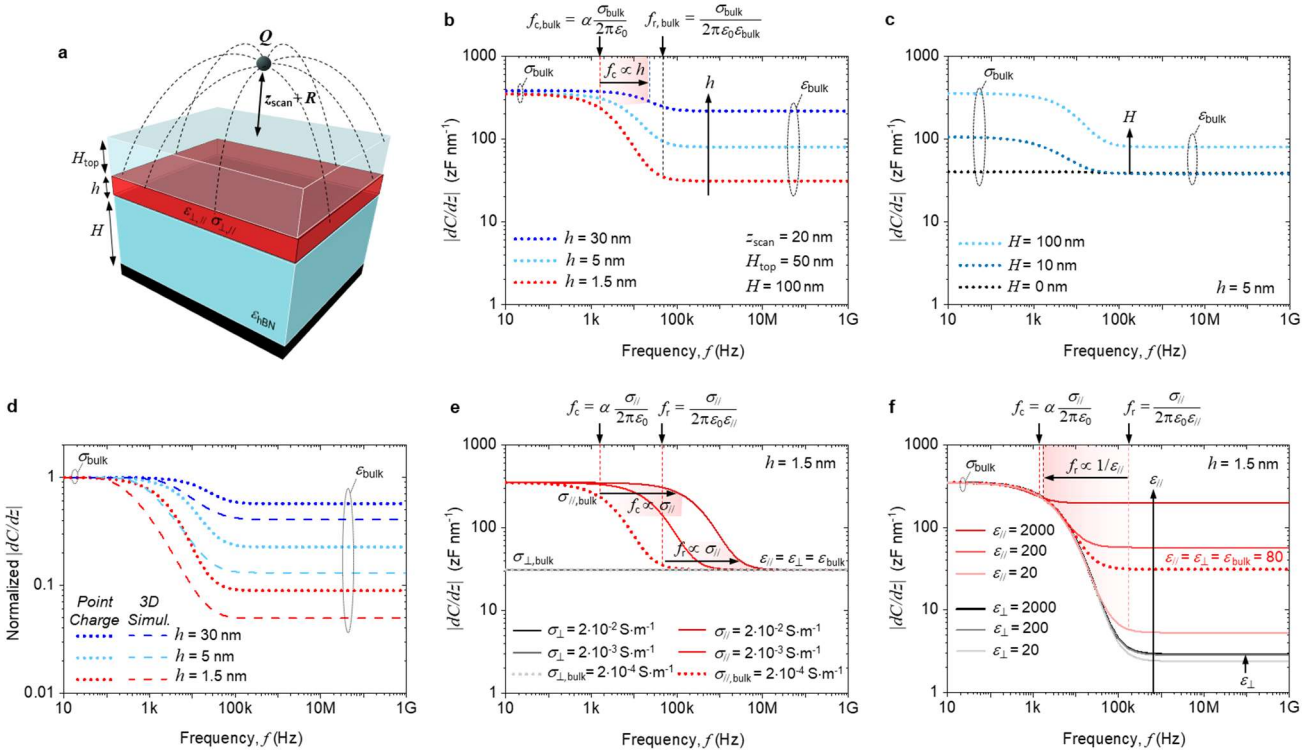


Extended Data Fig. 6 | Calculated dielectric spectra using the 3D numerical simulations. (**a**) Simplified illustration of our numerical model (shown in full in Extended Data Fig. 5). (**b**) Calculated $|dC/dz|$ as a function of frequency for different channel thicknesses (h = 30, 5 and 1.5 nm) and bulk water ($\varepsilon_{//} = \varepsilon_{\perp} = \varepsilon_{bulk}$, $\sigma_{//} = \sigma_{\perp} = \sigma_{bulk}$) inside the channels. (**c**) Same as in (b) for h = 5 nm but using various AFM tip radii and different channel widths. (**d**) Same as in (b,c) but for several w; normalized by the height of the low-f plateau. Dotted curve: corresponding calculations using our point-charge analytical model. (**e**) Same as (b) but for h = 1.5 nm and different water's $\sigma_{//}$ (red-colored curves, keeping $\sigma_{\perp}$ = 0,) and different $\sigma_{\perp}$ (grey-colored curves, $\sigma_{//}$ = 0). (**f**) Same as in (e) but for different $\varepsilon_{//}$ (red-colored curves, keeping $\varepsilon_{\perp}$ = 2) and different $\varepsilon_{\perp}$ (grey-colored curves, $\varepsilon_{//}$ = 2). The red dashed curve is the case of the channel filled with bulk water, same as shown in (b). Other parameters used in (b-f) (unless stated otherwise in the legends): R = 100 nm, θ = 25°, H_{cone} = 6 µm, $H_{cantilever}$ = 3 µm, $L_{cantilever}$ = 0 µm, z_{scan} = 20 nm; l = 3 µm, w_c = 200 nm, w_s = 800 nm, H_{top} = 50 nm and H = 100 nm. All the plotted $|dC/dz|$ curves are after subtracting the $|dC/dz|$ values found at the center of the hBN spacer region.



Extended Data Fig. 7 | Sensitivity to water's in-plane and out-of-plane dielectric constants and calibration curves using the 3D numerical simulations. (**a,b**) Calculated $|dC/dz|$ at 1 GHz as a function of water's $\varepsilon_{//}$ and $\varepsilon_{\perp}$ (with $\sigma_{//} = \sigma_{\perp} = \sigma_{\text{bulk}}$). h = 5 nm and the bottom hBN thicknesses (a) H = 0 (metallic substrate) and (b) H = 200 nm. (**c**) Calculated $|dC/dz|$ as in (a) and (b) but as a function of H for different combinations of $\varepsilon_{//}$ and $\varepsilon_{\perp}$. AFM cone height H_{cone} = 12 µm and cantilever length $L_{\text{cantilever}}$ = 20 µm (nominal parameters) to include long-range range effects that play a role for very thick H. (**d**) Cross-sections from (a) as a function of water's $\varepsilon_{\perp}$ for different values of $\varepsilon_{//}$. (**e**) Cross-sections from (b) as a function of $\varepsilon_{//}$ for different $\varepsilon_{\perp}$. (**f**) Calculated $|dC/dz|$ vs tip-surface distance above the center of the hBN spacer, as illustrated in the inset, for empty channels (black dashed line) and bulk-water-filled channels (cyan colored symbols) at various frequencies. The channel thickness is h = 5 nm and the bottom hBN thickness is H = 100 nm. Upper panel: error between the calculated curves for water-filled channels and that for empty channels. Other parameters used in (a-f) (unless stated otherwise in the legends): R = 100 nm, θ = 25°, H_{cone} = 6 µm, $H_{\text{cantilever}}$ = 3 µm, $L_{\text{cantilever}}$ = 0 µm, z_{scan} = 20 nm; l = 3 µm, w_c = 200 nm, w_s = 800 nm, H_{top} = 50 nm. All the plotted $|dC/dz|$ curves in panels (a-e) are after subtracting the $|dC/dz|$ values found at the center of the hBN spacer region.



Extended Data Fig. 8| Calculated dielectric spectra using the point-charge analytical model. (**a**) Its schematic. A point charge is used to approximate the AFM tip. The device is modeled as a water layer of infinite lateral size (red), which is encapsulated between top and bottom hBN layers (blue). For simplicity, the dielectric constant of hBN is assumed to be isotropic and equal to 4. (**b**) Calculated $|dC/dz|$ as a function of frequency for h = 30, 5 and 1.5 nm (color coded) and bulk water ($\varepsilon_{//} = \varepsilon_\perp = \varepsilon_{bulk}$, $\sigma_{//} = \sigma_\perp = \sigma_{bulk}$). (**c**) Same as in (b) for h = 5 nm but using various thicknesses of bottom hBN. (**d**) The dotted curves are the same as in (b) but normalized with respect to the low-f plateau's height. Dashed curves: corresponding analysis using the full 3D numerical simulations. (**e**) Same as in (b) for h = 1.5 nm and different $\sigma_{//}$ (red curves, keeping $\sigma_\perp = 0$) and $\sigma_\perp$ (grey, $\sigma_{//} = 0$). (**f**) Same as in (b) for h = 1.5 nm and various $\varepsilon_{//}$ (red solid curves, keeping $\varepsilon_\perp = 2$) and various $\varepsilon_\perp$ (grey curves, $\varepsilon_{//} = 2$). The red dotted curve is for the same channel filled with bulk water [also shown in (b)]. Other parameters used in (b-f) (unless stated otherwise in the legends) are z_{scan} = 20 nm, H_{top} = 50 nm and H = 100 nm. All the plotted $|dC/dz|$ curves are after subtracting the $|dC/dz|$ values found for the same geometry but with hBN instead of the water layer.